Optimization HW-4

(1.1) Downward Spiral

P_T for one MALA step at temperature $T > 0$.
All measurable sets are subsets of K.

For arbitrary probability measures, including point-mass initialization use $\|\mu - \nu\|_{TV} = \sup_A |\mu(A) - \nu(A)|$

(1) Include rejection probability
Let

$$\gamma_T(x) = 1 - \int_K k_T(x,y)\,dy$$

Integral is probability of proposing & accepting a point in K.
All remaining outcomes stay at x

$$P_T(x,A) = \int_A k_T(x,y)\,dy + \gamma_T(x)\,1_A(x)$$

Includes proposals outside K and proposals rejected by the acceptance test.

(2) Stationarity of the Gibbs distribution.
Stationarity equation is

$$\pi_T P_T = \pi_T, \quad \int_K P_T(x,A)\,\pi_T(dx) = \pi_T(A) \quad \text{for every } A$$

Accepted density × Gibbs density

$$p_T(x)\cdot k_T(x,y) = \min\{p_T(x)\,q_T(x,y),\ p_T(y)\,q_T(y,x)\}$$
$$= p_T(y)\,k_T(y,x)$$

Thus, accepted moves satisfy detailed balance

Integrating in x, $\int_K p_T(x)\, k_T(x,y)\, dx = p_T(y)\cdot(1-r_T(y))$

The stayput term contributes $p_T(y)\, r_T(y)$, so sum is $p_T(y)$

(3) Gibbs mass concentrates near minimum

Positive Volume: Compactness & continuity give minimizer $x^* \in K$. continuity gives a neighborhood of x^* in which $f < f^* + \epsilon/2$.

Take a ball $B(z,r) \subset K$ which exists because K has non-empty interior. For sufficiently small $s > 0$, convexity gives

$$(1-s)x^* + sB(z,r) \subset K \cap \{f < f^* + \epsilon/2\} = A_\epsilon$$

The ball on left has radius $sr > 0$

Hence, $v_\epsilon = \mathrm{Vol}_d(A_\epsilon) > 0$, even if x^* is on the boundary

Probability bound: Bound numerator on B_ϵ and denominator on A_ϵ

$$\int_{B_\epsilon} e^{-f(x)/T}\, dx \le V_\epsilon e^{-(f^*+\epsilon)/T}, \quad Z_T \ge v_\epsilon e^{-(f^*+\epsilon/2)/T}$$

Dividing, $\Pi_T(B_\epsilon) \le \frac{V}{v_\epsilon} e^{-\epsilon/2T}$

④ A uniform lower bound on transition probabilities

Set $M = (D + hG)^2 / 4h$ so $c_M = \Delta + M$

For any $x, y \in K$, $\| y - x + h \nabla f(x) \| \leq D + hG$

same bound holds after exchanging x & y

$\therefore$ Both proposal densities are at least $4\pi h T^{-d/2} e^{-M/T}$

Using acceptance rule,

$$k_T(x,y) = \min \left\{ q_T(x,y), e^{-(f(y) - f(x))/T} q_T(y,x) \right\}$$

Since $f(y) - f(x) \leq \Delta$, both expressions in minimum are at least $(4\pi h T)^{-d/2} e^{-(M+\Delta)/T}$ Thus,

$$k_T(x,y) \geq \frac{\alpha_T}{V}, \quad P_T(x,A) \geq \alpha_T U_K(A)$$

Particularly, $0 < \alpha_T \leq 1$ because $P_T(x,K) = 1$

⑤ Total variation contracts at fixed temp.

The minorization allows the decomposition.

$$P_T(x,\cdot) = \alpha_T U_K(\cdot) + (1 - \alpha_T) R_T(x,\cdot)$$

where R_T is a transition kernel if $\alpha_T < 1$. The common uniform term cancels and transition kernels do not increase total variation.

$$\| \mu P_T - \nu P_T \|_{TV} = (1-\alpha_T) \| \mu R_T - \nu R_T \|_{TV}$$

$$\leq (1-\alpha_T) \| \mu - \nu \|_{TV}$$

If $\alpha_T = 1$, the kernel equals Π_x for every starting pt. and same bound holds with 0 on the right

(6) Consecutive Gibbs distribution changes slowly,

Let $\beta = 1/T$ $\tilde{\rho}_\beta$ for normalized density. Derivative (hint)

$$\int_K | \partial_\beta \tilde{\rho}_\beta(x) | dx = \mathbb{E}_\beta | f - \mathbb{E}_\beta f | \leq \Delta$$

because both f and its mean lie in $[f^*, f^* + \Delta]$.
Integrating derivative b/w two inverse temps yields

$$\| \pi_{1/\beta'} - \pi_{1/\beta} \|_{TV} \leq \frac{1}{2} \Delta | \beta' - \beta |$$

Here, $\beta_n = \log(n+2)/c$ Hence,

$$\delta_n \leq \frac{\Delta}{2c} \log \frac{n+2}{n+1} \leq \frac{\Delta}{2c(n+1)}$$

We use $\log(1+u) \leq u$

(7) contraction amounts have divergent sum,

Put $\rho = C_M/C < 1$ and $A = V(4\pi hc)^{-d/2} > 0$

By direct substitution $\alpha_n = A\left[\log(n+2)\right]^{d/2}(n+2)^{-\rho}$

For $n \geq 1$, $\log(n+2) > 1$ so, $\alpha_n \geq A(n+2)^{-\rho}$

The p series diverges for $\rho < 1$ and

$$\therefore \sum_{n=1}^{\infty} \alpha_n = \infty$$

(8) Temperature drift is smaller than the contraction

combining (6) & (7)

$$0 \leq \frac{\delta_n}{\alpha_n} \leq \frac{\Delta}{2cA} \frac{(n+2)^{\rho}}{(n+1)\left[\log(n+2)\right]^{d/2}} \to 0$$

because $\rho - 1 < 0$. Also covers $\Delta = 0$ when $\delta_n = 0$

(9) Recursion for tracking moving Gibbs law

At step n, conditional on $X_{n-1} = x$ algorithm uses exactly $P_{T_n}(x, \cdot)$

Total probability law gives $\mu_n = \mu_{n-1} P_{T_n}$

By stationarity of Π_n, contraction & triangle inequ

$$e_n = \| \mu_{n-1} P_{T_n} - \pi_n P_{T_n} \|_{TV}$$
$$\leq (1-\alpha_n) \| \mu_{n-1} - \pi_n \|_{TV}$$
$$\leq (1-\alpha_n)(e_{n-1} + \delta_n)$$
$$\leq (1-\alpha_n) e_{n-1} + \delta_n$$

(10) Convergence to global minimum in prob.

Fix any $\gamma > 0$, Part 8 gives N such that $\delta_n < \gamma \alpha_n$ for $n \geq N$

Thus, $e_n - \gamma \leq (1-\alpha_n)(e_{n-1} - \gamma)$

Taking +ve parts & iterating,

$$(e_n - \gamma)_+ \leq (e_{N-1} - \gamma)_+ \prod_{j=N}^{n} (1-\alpha_j) \leq (e_{N-1} - \gamma)_+ \exp\left(-\sum_{j=N}^{n} \alpha_j\right) \to 0$$

Part (7) proves last limit. $\therefore$ lim sup $e_n \leq \gamma$

Since $\gamma > 0$ is arbitrary and $e_n \geq 0$, we have $e_n \to 0$

finally from (8) & $T_n \to 0$ implies

$$\mathbb{P}(f(X_n) \geq f_* + \epsilon) = \mu_n(B_\epsilon) \leq e_n + \frac{V}{\nu_\epsilon} e^{-\epsilon/2T_n} \to 0$$

Consequently, $f(X_n) \xrightarrow{\mathbb{P}} f_*$ for every initial distribution supported on K. Objective value convergence.

AI Usage: ChatGPT 5o1 6.1 was used to understand the problem and help with solution suggestions checking, refining as needed for the final submission.

Implementation writeup

Exercise 2.1: Minibatch SGD

- **Minibatch evaluation:** vectorize the scalar loss over the fixed-capacity batch with `jax.vmap`. Index the sample PyTree by lane; the empty deterministic sample `()` also works.
- **Masking:** average only active lanes with the supplied `masked_mean`. Return all lane losses, including padded lanes, so the solver can compute its statistics.
- **Derivatives:** when individual gradients are requested, vectorize `jax.value_and_grad` and average the active gradients. Otherwise, differentiate the masked mean directly with `has_aux=True` to return the losses without storing individual gradients.
- **Momentum:** use the classical recurrence `v_new = momentum * v_old + gradient`. Return `v_new` as both the direction and the new `SGDState.velocity`. The outer solver applies the learning rate and projection.
- **Compatibility:** preserve PyTree structure, shapes, and floating dtypes. The implementation supports JIT, vectorization, and differentiation through the updates.

Experiment settings

- Use the supplied quadratic and simplex objectives, their original parameters, and their exact reference solutions.
- Quadratic: target `(0.8, -0.6, 0.3, 1.1)`, diagonal curvature `(0.7, 1.3, 2.2, 4.0)`, forcing scale `0.35`, initial point zero, and momentum `0.6` as in the supplied example.
- Simplex: marginal benefits `(0.70, 0.45, 0.15, -0.30)`, forcing scale `0.15`, uniform initial allocation, no momentum, and projection onto the unit simplex. The exact solution is `(0.60, 0.35, 0.05, 0)`.
- Sweep learning rates `{0.01, 0.1, 0.3, 0.9}` and batch sizes `{1, 16, 64}` for **400 iterations each**. Use ten independent seeds, `0,...,9`, in float64 on CPU.
- Reuse a seed across configurations to share random numbers where possible; the ten trials within each configuration are independent.
- Plot the mean **true expected objective gap**, rather than noisy minibatch loss. Error bars are one sample standard deviation across the ten trials, drawn at selected iterations.
- The top row uses iteration count; the bottom row uses actual sample count `batch_size * iteration`. Padding is not counted as data.
- Lower error bars are truncated to remain positive on the logarithmic axis. The CSV retains the full standard deviations. Standard deviations for the deliberately divergent run are computed after scaling to avoid numerical overflow.

Figure 1: Quadratic learning-rate and batch-size sweep (see figures at the end).

Figure 2: Simplex learning-rate and batch-size sweep (see figures at the end).

Which hyperparameters perform best?

- **After 400 iterations, quadratic:** learning rate `0.01`, batch `64` gives the lowest observed mean gap, **4.6543e-5**, with run SD **2.8843e-5**. It uses **25,600 samples**.
- **After 400 iterations, simplex:** learning rate `0.1`, batch `64` gives the lowest observed mean gap, **1.6807e-5**, with run SD **2.0068e-5**. It also uses **25,600 samples**.
- **At an equal budget of 384 samples, quadratic:** learning rate `0.1`, batch `16` gives the lowest observed mean gap, **1.3021e-3**, after **24 iterations**.
- **At the same 384-sample budget, simplex:** learning rate `0.01`, batch `1` gives the lowest observed mean gap, **1.1187e-4**, after **384 iterations**.

- These are empirical rankings within the tested grid and budgets. The overlapping run variability does not establish a statistically unique best setting.
- The complete 24-configuration results are in sgd_summary.csv. All ten gap curves for each configuration are saved in sgd_gaps.npz.

Notable behavior

- **Noise floor:** stable constant learning rates approach a fluctuating neighborhood of the optimum. Larger batches reduce this fluctuation; smaller learning rates usually give lower late-stage error.
- **Sample cost:** a large batch can win at a fixed iteration count but use many more samples. With few samples available, fewer large-batch updates can leave substantial initial error.
- **Quadratic instability:** learning rate `0.9` diverges for every batch size. With momentum `m=0.6`, a scalar curvature mode is stable only when `0 < learning_rate * curvature < 2(1+m)`. Since the maximum curvature is `4`, the bound is `learning_rate < 0.8`; `0.9` violates it. Increasing the batch size cannot repair this deterministic instability.
- **Simplex feasibility:** every recorded allocation is nonnegative and sums to one within numerical precision. Projection keeps the iterates feasible even when the stochastic direction points outside the simplex.
- **Boundary solution:** the fourth allocation is zero at the optimum; random gradients can activate small positive allocations, which projection then corrects.

Exercise 2.2: Langevin methods

- **Score convention:** for `score = gradient(log p_T) = -gradient(energy)/T`, return `-T * score` to the optimizer. This produces the same gradient as the energy formulation.
- **Temperature timing:** evaluate the score and noise temperature policies using the before-step context, which contains the previous minibatch statistics and old optimizer state.
- **Noise:** perturb the deterministic candidate by `sqrt(2 * learning_rate * temperature) * z`. Split the reserved PRNG key across PyTree leaves; sample every entry independently in the leaf's dtype.
- **Update order:** deterministic optimizer step, then Langevin perturbation, then projection. Objective sampling and Langevin noise use separate random streams supplied by the solver.
- **Diagnostics:** return evaluated temperature, noise scale, the global norm of the actual perturbation, and the temperature-validity flag.
- **Corner cases:** zero temperature gives the deterministic candidate exactly when allowed. Invalid constant temperatures raise at construction; invalid numeric dynamic temperatures produce NaNs and a false validity flag. Wrong shapes or non-real dtypes raise through the supplied helpers.
- **Differentiability:** retain continuous derivatives through scores, temperatures, and arguments. Use the supplied `safe_sqrt`, whose derivative at zero is defined to be zero.

Harmonic energy/score comparison

- Repeat the supplied 64-step forced harmonic problem with ten seeds. Use the same key for the energy and score versions within each trial.
- Keep the supplied learning rate `0.04`, temperature schedule from `0.75` to `0.30`, and batch schedule from `4` to `19`.
- Across all ten seeds and all coordinates, the largest path difference is **4.4409e-16**. The descriptions agree to floating-point precision.
- The plot overlays both descriptions for three representative seeds. The dashed score curves lie on the energy curves. All ten paired trajectories and discrepancies are saved in harmonic_paths.npz.

Figure 3: Shared-key harmonic trajectories (see figures at the end).

Gaussian ensemble

- Repeat the supplied experiment with **ten independent ensembles**, each containing **4,096 independent chains** and **18 updates**. Use initial state `-1.8`, mean `0.7`, curvature `1.4`, learning rate `0.07`, and temperature `0.7`.
- The left panel shows ten representative chains from the first ensemble. The middle panel plots the mean across ensemble means; its error bars are nominal 95% Student-t intervals, `2.262 * SD / sqrt(10)`. The right panel gives the analogous variance comparison.
- The final mean across ensembles is **0.308501**; the SD of the ten ensemble means is **0.010508**. The mean final within-ensemble variance is **0.511077**.
- This 18-step run has **not reached equilibrium**. The exact discrete recursion has contraction `a = 1 - 0.07 * 1.4 = 0.902`, mean `0.7 + a^k * (-1.8 - 0.7)`, and variance `2 * 0.07 * 0.7 * (1 - a^(2k)) / (1 - a^2)`.
- At step 18, these formulas predict mean **0.309467** and variance **0.512932**, consistent with the measurements.
- Even after full mixing, this unadjusted finite-step method has stationary variance **0.525762**, while the continuous Gibbs target has variance **0.5**. The discretization introduces bias; there is no Metropolis correction in this code.
- Saved ensemble means, variances, analytical curves, and representative chains are in gaussian_ensembles.npz.

Figure 4: Gaussian chains and repeated-ensemble error bars (see figures at the end).

Validation and reproduction

- The **11 passing tests** cover masked batches in float32/float64, nested parameter trees, empty samples, higher derivatives, momentum, score conversion, invalid temperatures, independent noise, diagnostics, exact zero-temperature behavior, projection order, and continuation.
- The supplied stochastic and Langevin example programs also run successfully. Their complete reports are stochastic_examples.json and langevin_examples.json.
- The supplied Langevin examples also pass in float32; the shared-key harmonic discrepancy is `2.98e-8`, consistent with its lower precision. The report is langevin_examples_float32.json.
- From the submission root, run `bash setup.sh`, then `.venv-implementation/bin/python -m pytest Implementation/tests -q`.
- Regenerate the assigned plots with `MPLCONFIGDIR=/tmp/hw4-matplotlib .venv-implementation/bin/python Implementation/run_experiments.py`.
- See the root README for all commands, package versions, and output locations.

ChatGPT was used to understand the problems, assist with equations and implementation, check the results, and prepare this writeup.

Application writeup

Exercise 3.1: Chance-constrained geometry compensation

- Optimize dimensionless commands `x`; convert them to millimeter commands using `u = Sx`.
- Use the supplied affine error model `a = b + GSx + Lξ`, with the full supplied covariance `Σ = LLᵀ`.
- Set the solver response to **H = GS**. Keep the supplied bias, covariance, case name, and requested probability unchanged.
- Enforce **one joint event**: every output coefficient must satisfy its interval simultaneously with probability at least `alpha`. Separate marginal 95% constraints would not imply 95% joint success.
- Return the formulation in `AffineGaussianBox`; use the supplied probability oracle and Veinott solver afterward.

DLP and CNC formulations

- Let `d` be the number of supplied coefficients, `W` the vertex interpolation matrix, and `ω` the normalized vertex-area weights.
- Since the surface penalty is `0.5 * [0.85 * sum_i ω_i (Wx)_i² + (0.15/d) * ||x||²]`, the solver data are **Q = 0.85 Wᵀ diag(ω) W + (0.15/d) I** and **c = 0**.
- Set `S = tolerance_mm * I`, so **H = tolerance_mm * G**. The objective is dimensionless as specified; do not insert an extra squared millimeter scale into `Q`.
- Set all coefficient limits to `[-tolerance_mm, +tolerance_mm]`.
- **DLP:** use decision bounds `[-4, 4]`. The tolerances are `0.200 mm` for the ellipsoid and `0.160 mm` for the sleeve.
- **CNC:** use decision bounds `[-4, 0]` to permit only inward compensation. The tolerances are `0.120 mm` for the cylinder and `0.140 mm` for the pocket plate.
- A shared private helper assembles this common surface formulation; only the decision upper bound differs between DLP and CNC.

Embedded extrusion formulation

- Read the control count `K` and section count `N_s` from `basis.weights.shape`; preserve channel order `(dx, dy, dz, radius)`.
- Build the half-width vector **t = ((τ_p/√3) 1_(3K), τ_r 1_K)**. Set **S = diag(t)**, **H = GS**, **lower = -t**, and **upper = t**.
- Use the supplied budgets and radius. In these cases `τ_p = 0.180 mm`, `τ_r = 0.060 mm`, and `r_0 = 0.600 mm`.
- With `X` formed by putting the four channel vectors in columns, define **M = (0.85/N_s) WᵀW + (0.08/K) I + (0.07/K) D₂ᵀD₂**.
- The requested objective is `(1/8) * sum_channels x_channelᵀ M x_channel`. Matching the solver's `(1/2) xᵀQx` gives **Q = block_diag(M, M, M, M) / 4**, **c = 0**.
- Form `D₂` from second differences of the identity. This also handles fewer than three controls, when the smoothness term has no rows.
- Set centerline decision bounds to `[-6, 6]` and radius bounds to `±0.45 r_0 / τ_r`, which are `±4.5` for the supplied data.
- For each section and XYZ coordinate, use `A_pos x = (τ_p/√3) W x_channel`. Append a zero block for radius controls.
- Reserve the maximum allowed radius `1.45 r_0` at each bath wall. Stack **A_ub = [A_pos; -A_pos]** and **b_ub = [B_max - 1.45 r_0 - p*; p* - B_min - 1.45 r_0]**, with coordinates stacked in channel order.

- These inequalities keep the commanded centerline and the reserved circular filament envelope inside the bath at every supplied path section.

Why the coefficient event controls geometry

- DLP/CNC basis weights are nonnegative and sum to one. If every output coefficient has magnitude at most τ, every interpolated normal displacement also has magnitude at most τ.
- For embedded extrusion, each interpolated centerline component is bounded by $\tau_p/\sqrt{3}$, so the centerline displacement norm is at most τ_p. The radius displacement is at most τ_r.
- The triangle inequality then bounds each embedded ring-vertex displacement by $\tau_p + \tau_r = 0.240$ mm. The target ring directions remain fixed.
- The coefficient-box event is sufficient for the mesh tolerance. It can be conservative, so directly simulated whole-mesh yield can exceed coefficient-box yield.

Numerical procedure and validation

- Run all six physical cases at $\alpha \in \{0.90, 0.95, 0.99\}$ in separate directories. The `--case all` command also runs the three supplied abstract checks.
- Use the supplied settings: `2,048` optimizer QMC points, a `0.004` probability buffer, and `100,000` independent validation draws per physical case.
- The working probability targets are `0.904`, `0.954`, and `0.994`. The buffer provides numerical headroom; it does not certify integration accuracy.
- Check each solution with four independent `8,192`-point QMC scrambles and the supplied SciPy Gaussian-box integral.
- Report individual two-sided 95% Clopper-Pearson intervals for the Monte Carlo yields. These intervals cover sampling error, not uncertainty in the process model.
- All objective and bath-constraint formulas pass **six tests**, including embedded problems with `2`, `3`, `5`, and `6` controls.
- The process models are the assignment's synthetic surrogates. The diagnostic LP gap from approximate probability cuts is not a certified global optimality bound.

Reproduce the application

- From the submission root, first run `bash setup.sh`.
- Run `.venv-application/bin/python -m geocomp --case all --alpha 0.90 --out Application/results_a090 --verbose`.
- Repeat with `--alpha 0.95 --out Application/results_a095` and `--alpha 0.99 --out Application/results_a099`.
- Run `.venv-application/bin/python Application/summarize_results.py` to regenerate the tables, static figures, and the results below.
- Test the formulas with `.venv-application/bin/python -m pytest Application/tests -q`.

Results from all 18 physical runs

- All **18 physical cases converged**. The unrounded optimized decisions and millimeter commands are in optimized_variables.csv.
- The table gives independent Monte Carlo coefficient-box yield and its individual 95% interval; the whole-mesh yield is listed separately.
- Full precision probability checks, model data, and solver traces are retained in each case's `metrics.json`, `arrays.npz`, and `optimization.json`.

Case	alpha	Cost	Iterations	Box yield [95% CI]	Mesh yield
dlp_ellipsoid	0.90	0.168830	69	0.90326 [0.90141, 0.90509]	0.93580
dlp_sleeve	0.90	0.301998	81	0.90336 [0.90151, 0.90518]	0.92759
cnc_cylinder	0.90	0.537028	85	0.90450 [0.90266, 0.90631]	0.92396
cnc_pocket_plate	0.90	0.336961	85	0.90459 [0.90275, 0.90640]	0.90639
embedded_helix	0.90	0.762813	310	0.90483 [0.90299, 0.90664]	1.00000
embedded_serpentine	0.90	0.594633	330	0.90489 [0.90305, 0.90670]	1.00000
dlp_ellipsoid	0.95	0.189690	68	0.95425 [0.95294, 0.95554]	0.97174
dlp_sleeve	0.95	0.330274	78	0.95403 [0.95271, 0.95532]	0.96633
cnc_cylinder	0.95	0.579801	82	0.95449 [0.95318, 0.95577]	0.96613
cnc_pocket_plate	0.95	0.369622	81	0.95513 [0.95383, 0.95640]	0.95615
embedded_helix	0.95	0.802153	301	0.95471 [0.95340, 0.95599]	1.00000
embedded_serpentine	0.95	0.632158	319	0.95484 [0.95353, 0.95612]	1.00000
dlp_ellipsoid	0.99	0.244898	73	0.99399 [0.99349, 0.99446]	0.99684
dlp_sleeve	0.99	0.402146	80	0.99407 [0.99357, 0.99454]	0.99600
cnc_cylinder	0.99	0.685798	89	0.99404 [0.99354, 0.99451]	0.99602
cnc_pocket_plate	0.99	0.452076	80	0.99383 [0.99333, 0.99431]	0.99409
embedded_helix	0.99	0.900518	300	0.99419 [0.99370, 0.99465]	1.00000
embedded_serpentine	0.99	0.726903	340	0.99448 [0.99400, 0.99493]	1.00000

Interpretation

- Every optimized coefficient-box Monte Carlo lower confidence limit exceeds its requested target. These are individual intervals, not a simultaneous coverage statement over all 18 experiments.
- The smallest lower-limit headroom is **0.001411** in probability units.
- The largest difference between the optimizer probability and the independent QMC mean is **3.317e-05**; the corresponding largest SciPy difference is **3.995e-05**.
- Compensation cost increases with the requested yield for every geometry: tighter chance constraints shrink the feasible set.
- The uncompensated coefficient-box probabilities are extremely small. Correcting systematic bias produces a large improvement in modeled yield.
- The embedded whole-mesh yield is especially conservative relative to the coefficient box: the sum of separate centerline and radius budgets bounds displacement, but both worst cases need not occur together.

Figure 5: Validated coefficient and mesh yields (see figures at the end).

Figure 6: Compensation cost against required yield (see figures at the end).

Exaggerated geometry visualizations

- DLP/CNC static figures use **8× displacement**; embedded figures use **4× displacement**. Gray is the target and blue is the compensated command.
- The supplied offline viewers include target, command, uncompensated output, compensated output, paired random realizations, bath context, and solver convergence. Select the display factor and realization using their controls.
- Exported STLs retain the actual millimeter coordinates. Exaggeration only changes the display.
- **alpha = 0.90:** Open the supplied interactive viewer.

Figure 7: Exaggerated commands at alpha 0.90 (see figures at the end).

- **alpha = 0.95:** Open the supplied interactive viewer.

Figure 8: Exaggerated commands at alpha 0.95 (see figures at the end).

- **alpha = 0.99:** Open the supplied interactive viewer.

Figure 9: Exaggerated commands at alpha 0.99 (see figures at the end).

Optimized dimensionless variables

- Entries below are rounded to six decimals for reading; use the linked metrics or CSV for all available digits.
- Surface vectors follow the supplied mode order. Embedded vectors are shown in four channel blocks, each ordered along the path.

alpha = 0.90

- **dlp_ellipsoid** (full precision):
  - `x = (1.268171, 1.427831, 0.448316, 0.431907, -0.157148, -0.182068, 0.456970, 0.449055)`
- **dlp_sleeve** (full precision):
  - `x = (0.754516, 1.524176, 0.772195, 1.462838, 0.100713, 0.093954, -0.868357, -0.873380)`
- **cnc_cylinder** (full precision):
  - `x = (-1.231708, -0.873973, -1.216898, -0.838807, -1.208710, -1.205373, -0.529888, -0.972368)`
- **cnc_pocket_plate** (full precision):
  - `x = (-0.943201, -0.624061, -0.326392, -0.304179, -1.076170, -1.306351, -1.273435, -0.953671)`
- **embedded_helix** (full precision):
  - `dx = (0.999990, 1.107560, -4.316296, 1.093960, 1.379306)`
  - `dy = (-0.108310, 2.091281, -0.017354, -2.492548, 0.214857)`
  - `dz = (0.463420, 0.451143, 0.499268, 0.545061, 0.598919)`
  - `radius = (-0.773001, -0.771026, -0.833999, -0.901544, -0.965721)`
- **embedded_serpentine** (full precision):
  - `dx = (0.257257, 1.391486, -0.112517, 1.613321, 0.457377)`
  - `dy = (0.814478, 1.017484, -3.791354, 1.023499, 1.086926)`
  - `dz = (-0.008026, 0.561767, 0.521763, 0.663409, 0.010107)`
  - `radius = (-0.767863, -0.767730, -0.831758, -0.900846, -0.962647)`

alpha = 0.95

- **dlp_ellipsoid** (full precision):
  - `x = (1.322988, 1.479731, 0.497772, 0.482141, -0.144748, -0.156181, 0.498998, 0.497649)`

- **dlp_sleeve** (full precision):
  - `x = (0.804237, 1.577802, 0.824900, 1.513931, 0.131964, 0.131730, -0.918141, -0.917996)`
- **cnc_cylinder** (full precision):
  - `x = (-1.272154, -0.914126, -1.257915, -0.878031, -1.253160, -1.251225, -0.566031, -1.011506)`
- **cnc_pocket_plate** (full precision):
  - `x = (-0.982945, -0.664226, -0.358649, -0.340633, -1.124797, -1.353081, -1.320062, -0.992890)`
- **embedded_helix** (full precision):
  - `dx = (1.030658, 1.155189, -4.376541, 1.143998, 1.421921)`
  - `dy = (-0.109258, 2.140464, -0.017059, -2.548545, 0.220858)`
  - `dz = (0.497444, 0.489800, 0.540297, 0.587307, 0.640493)`
  - `radius = (-0.807131, -0.808081, -0.877923, -0.944923, -1.011355)`
- **embedded_serpentine** (full precision):
  - `dx = (0.291039, 1.440971, -0.124358, 1.668987, 0.503817)`
  - `dy = (0.835396, 1.065144, -3.851629, 1.074487, 1.133757)`
  - `dz = (0.013629, 0.603672, 0.558398, 0.706718, 0.036665)`
  - `radius = (-0.804626, -0.808178, -0.874287, -0.942843, -1.006037)`

alpha = 0.99

- **dlp_ellipsoid** (full precision):
  - `x = (1.453583, 1.595464, 0.605213, 0.592703, -0.081416, -0.079828, 0.603070, 0.607965)`
- **dlp_sleeve** (full precision):
  - `x = (0.924719, 1.695752, 0.943543, 1.632900, 0.229156, 0.225136, -1.031325, -1.026161)`
- **cnc_cylinder** (full precision):
  - `x = (-1.369194, -1.003628, -1.351793, -0.968587, -1.358933, -1.357256, -0.653128, -1.100673)`
- **cnc_pocket_plate** (full precision):
  - `x = (-1.081307, -0.758971, -0.433950, -0.426704, -1.231012, -1.459886, -1.426813, -1.089575)`
- **embedded_helix** (full precision):
  - `dx = (1.105091, 1.263863, -4.513495, 1.264932, 1.526464)`
  - `dy = (-0.118504, 2.257699, -0.004182, -2.679126, 0.235757)`
  - `dz = (0.587667, 0.587507, 0.636947, 0.691001, 0.742586)`
  - `radius = (-0.897150, -0.903318, -0.977083, -1.049355, -1.112660)`
- **embedded_serpentine** (full precision):
  - `dx = (0.374742, 1.558226, -0.134202, 1.797848, 0.612242)`
  - `dy = (0.891603, 1.175599, -3.987815, 1.195366, 1.232573)`
  - `dz = (0.083931, 0.699410, 0.657618, 0.816557, 0.120636)`
  - `radius = (-0.895868, -0.903886, -0.974501, -1.048707, -1.107587)`

ChatGPT was used to understand the problems, assist with equations and implementation, check the results, and prepare this writeup.

Figures

Figure 1. Quadratic learning-rate and batch-size sweep

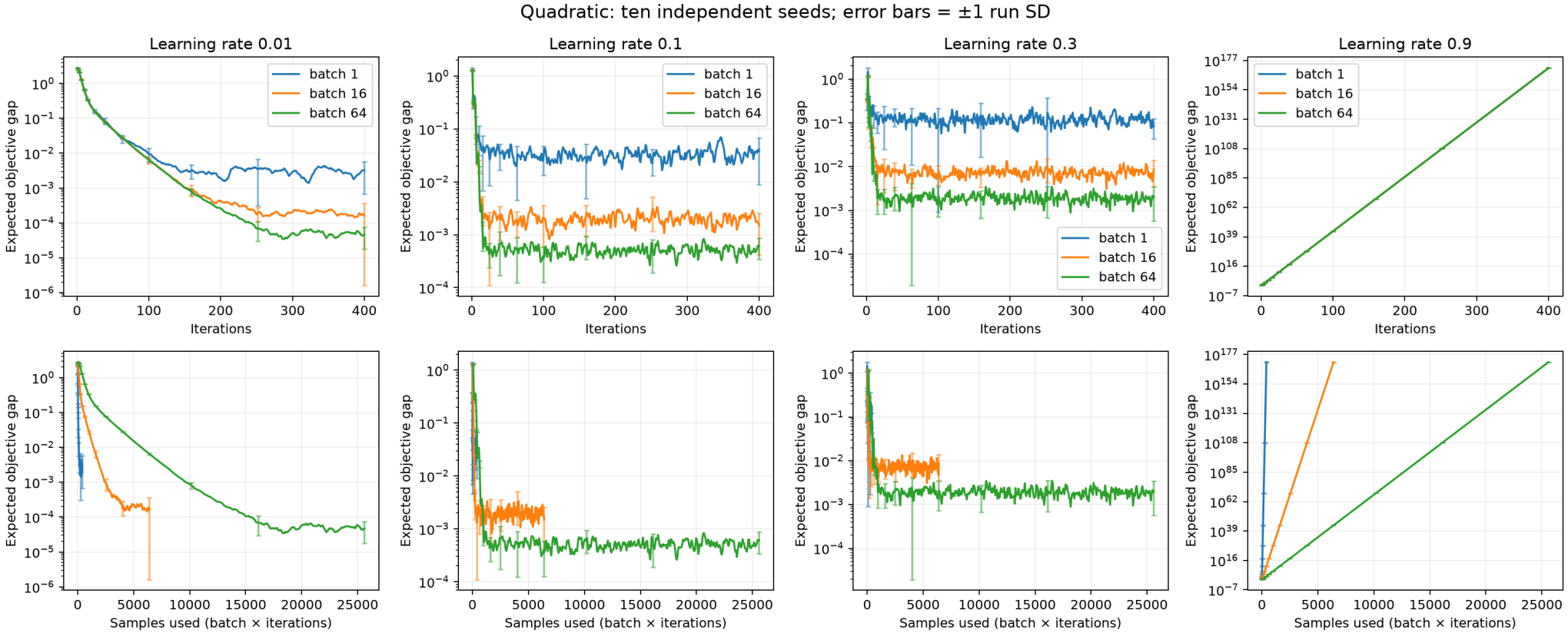


Figure 2. Simplex learning-rate and batch-size sweep

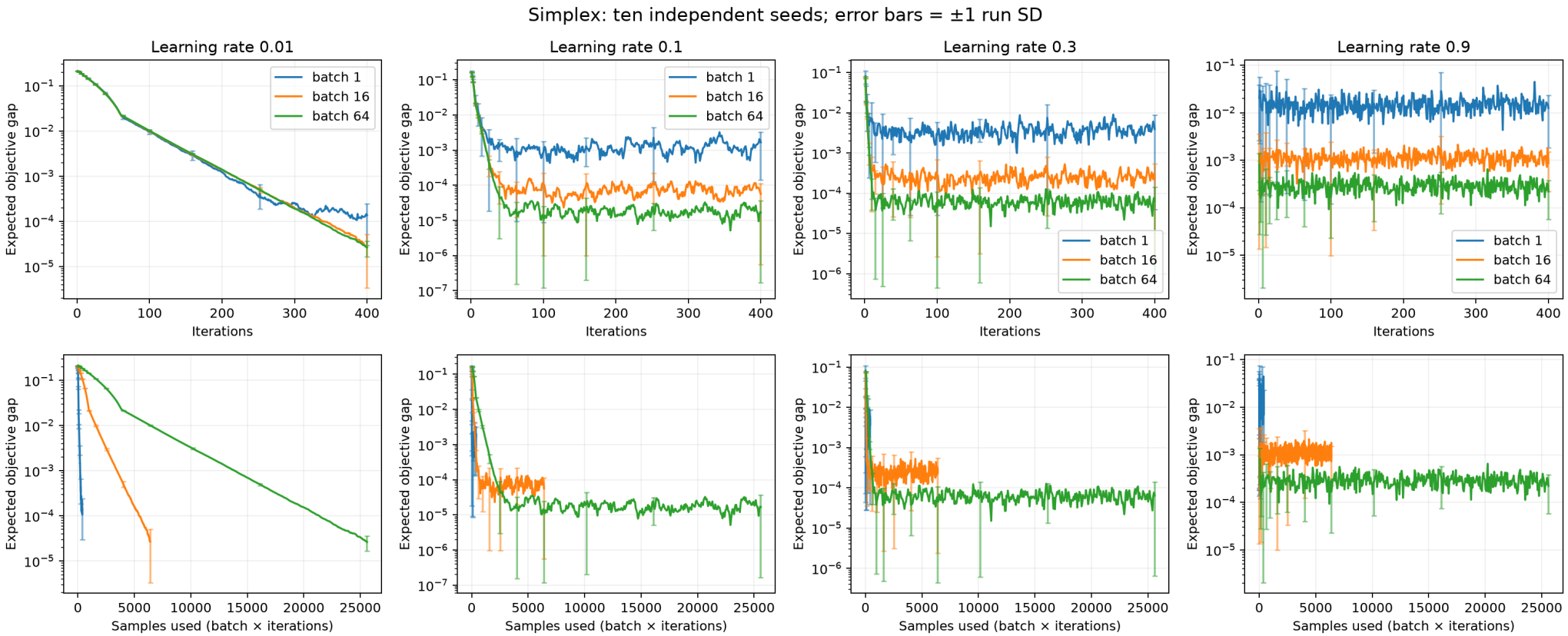


Figure 3. Shared-key harmonic trajectories

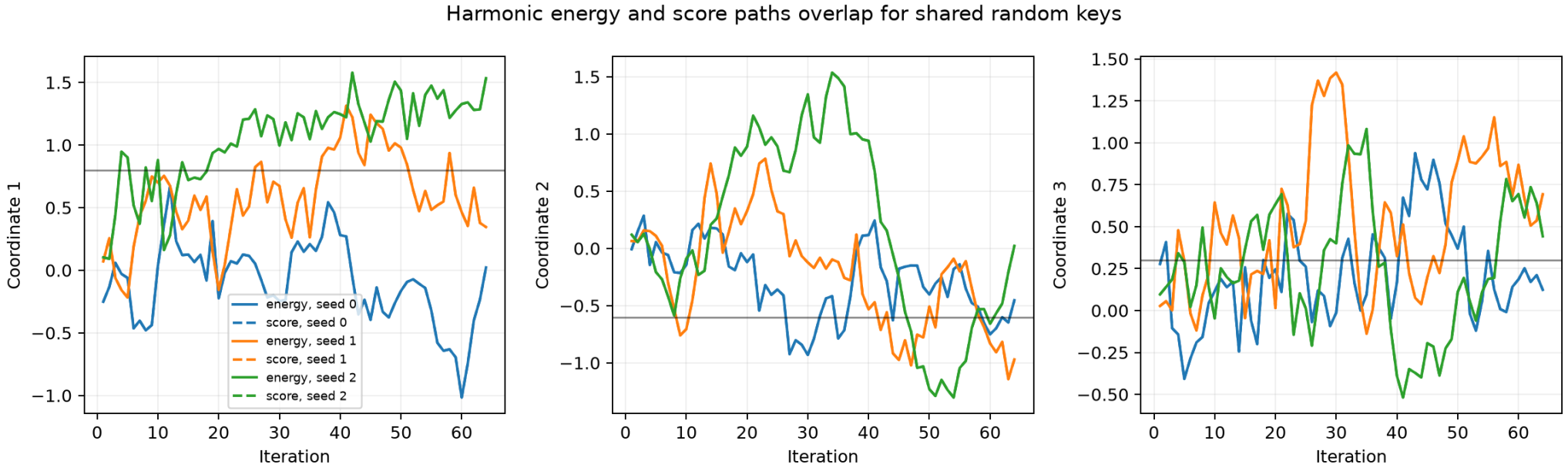


Figure 4. Gaussian chains and repeated-ensemble error bars

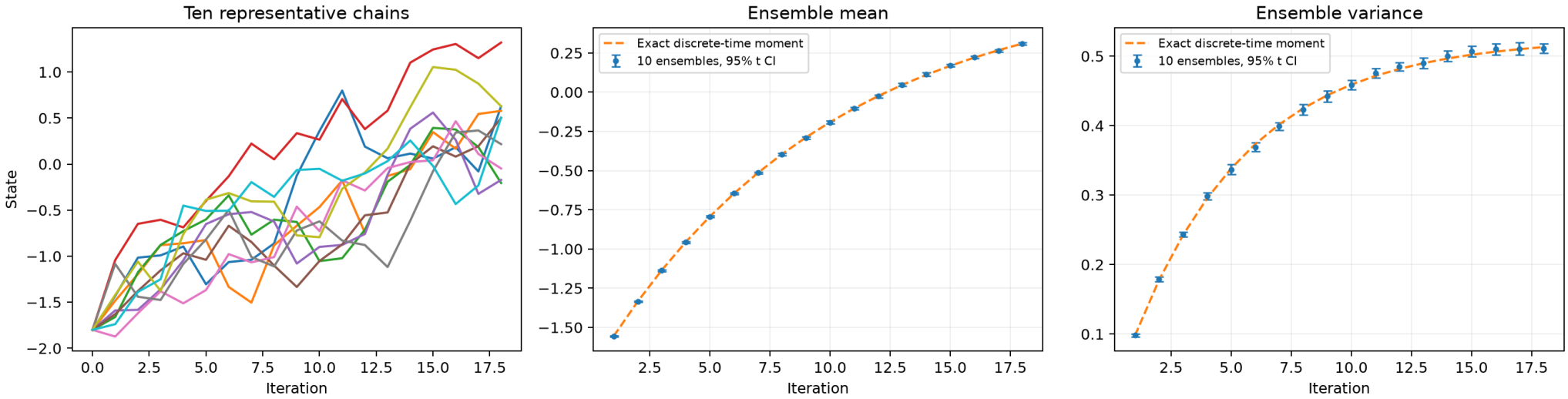


Figure 5. Validated coefficient and mesh yields

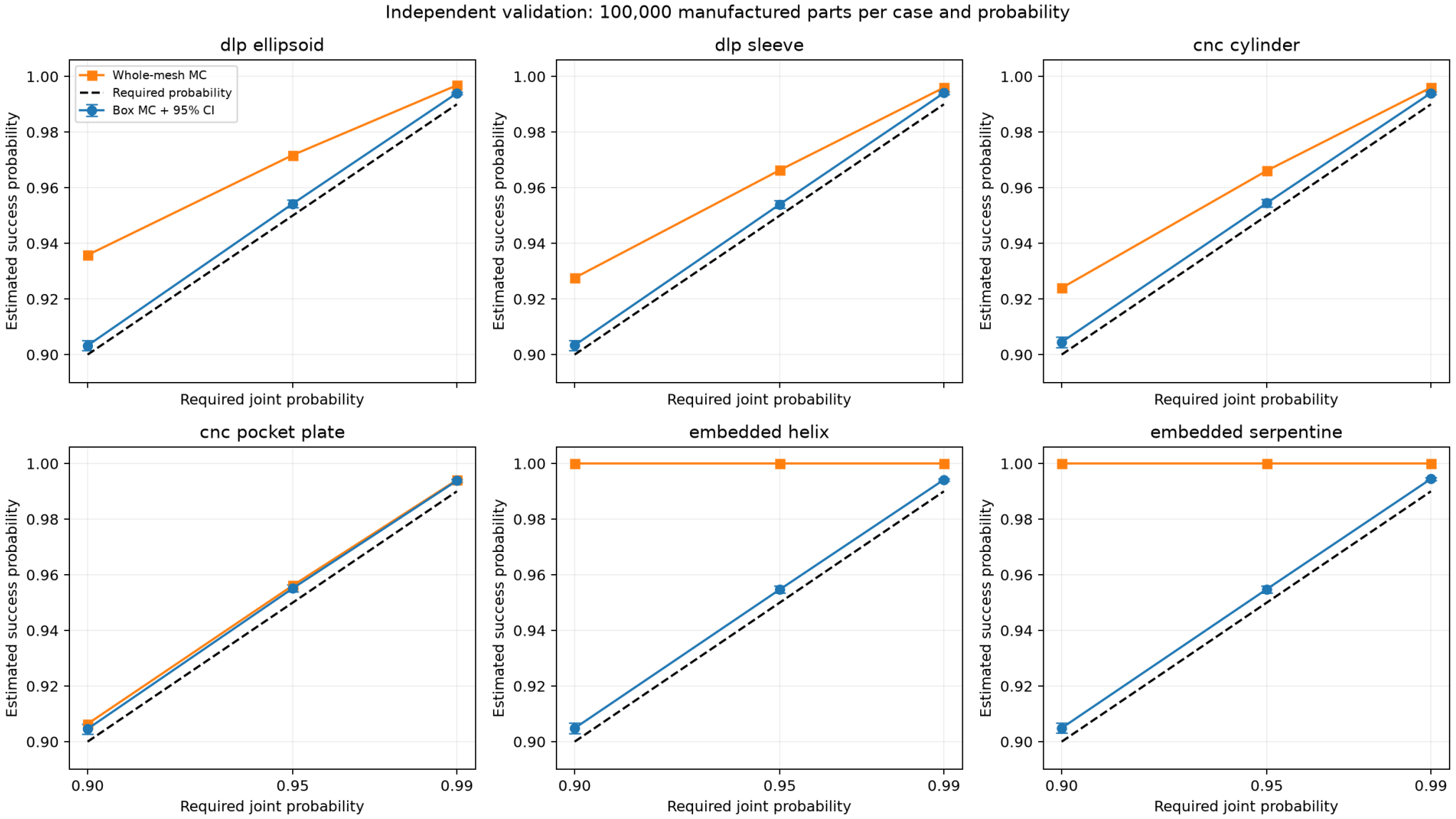


Figure 6. Compensation cost against required yield

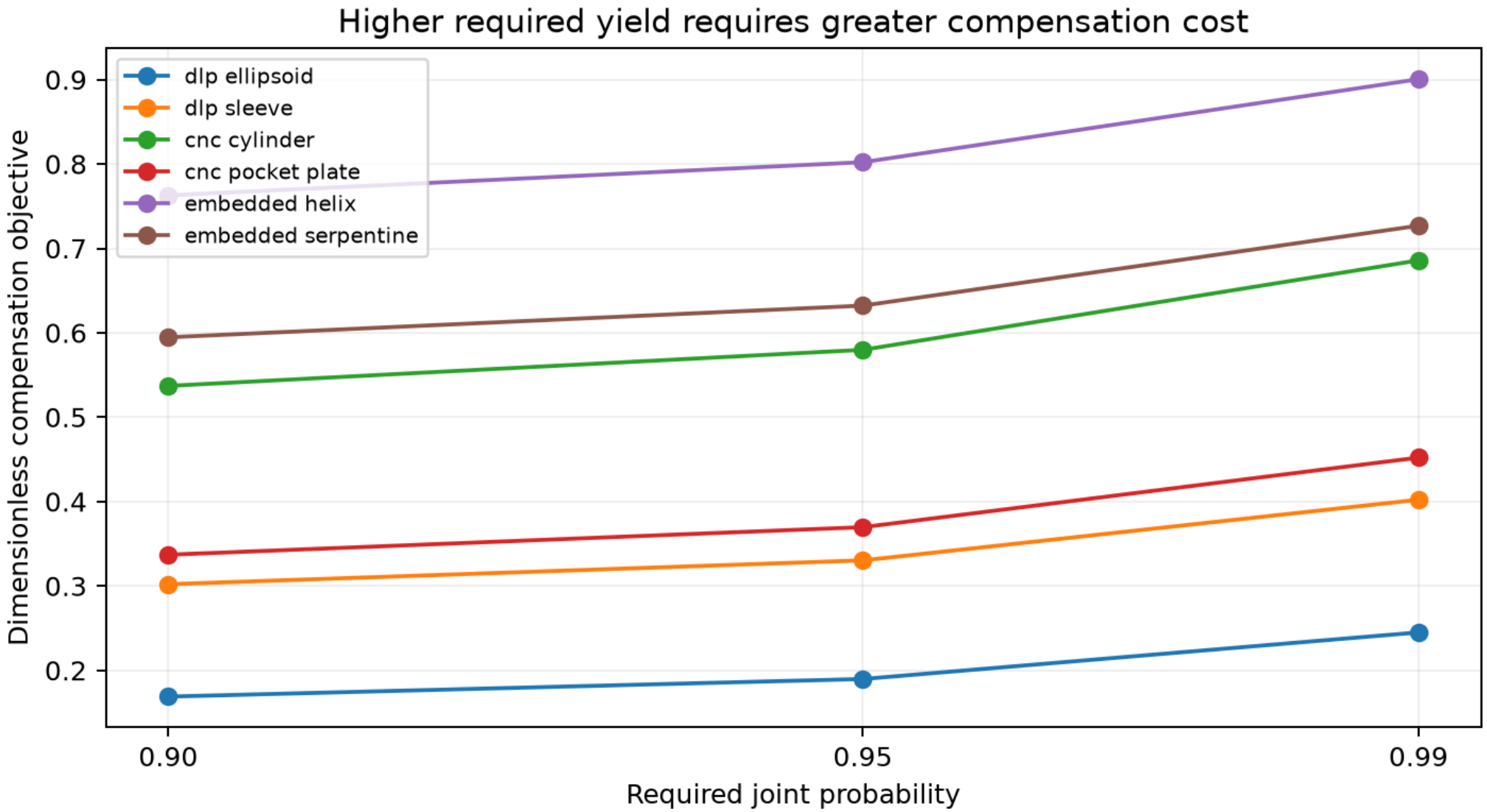


Figure 7. Exaggerated commands at alpha 0.90

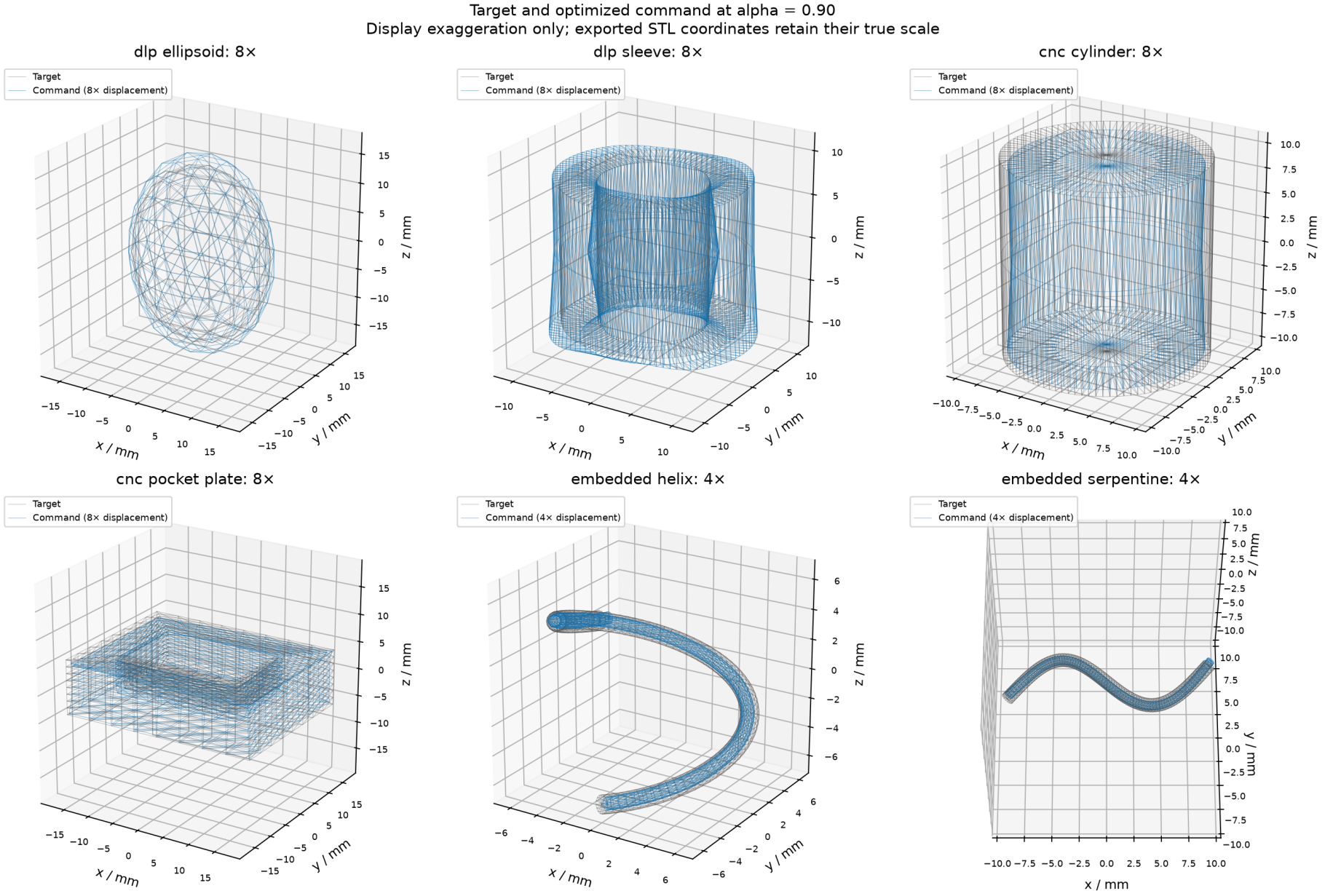


Figure 8. Exaggerated commands at alpha 0.95

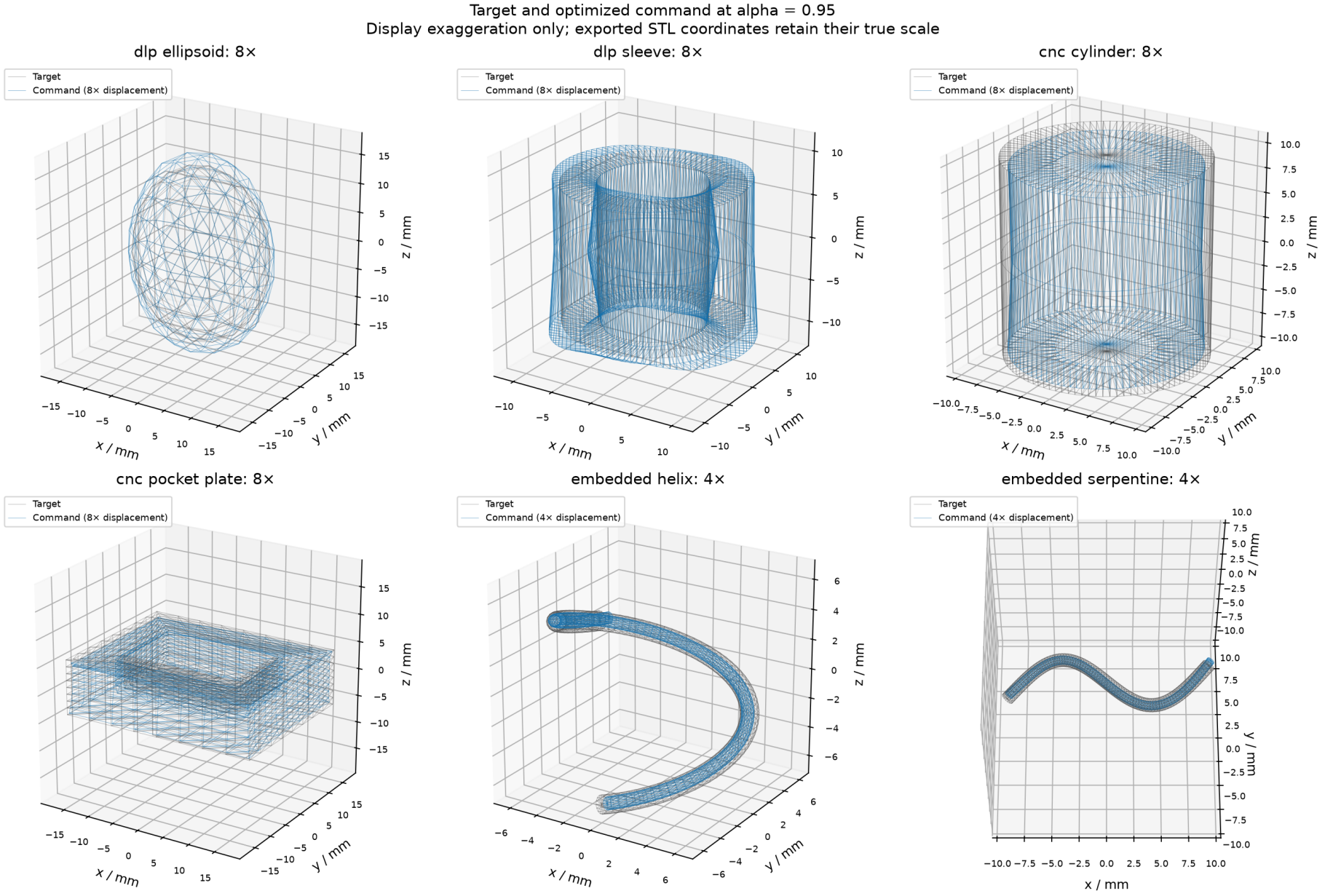


Figure 9. Exaggerated commands at alpha 0.99

